

Toward Coordinated-Voltage-Control-Enabled HV Smart Grids

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Abstract—In this paper we propose a Coordinated Voltage Control-Enabled HV Smart Grid framework (CVC, in short) for utilizing voltage-controllable equipment in support of enhanced reliable and efficient operations. In the first part of the paper we present the basic problem formulation underlying CVC. This is followed by assessing the effects of performance objectives selected, the type of controllable equipment used and the protocols for allocating responsibilities among Independent System Operators (ISOs) and the equipment owners (both Transmission Owners (TOs) and Generation Owners (GOs)). In the second part of the paper we describe potential benefits from the proposed CVC based on the large-scale simulations for the New England (NE), New York (NY) and Pennsylvania-Jersey-Maryland (PJM) power systems. The simulations for NE and NY power systems are carried out using an AC Extended Optimal Power Flow (AC XOPF), instead of commonly used DC OPF. The use of AC XOPF is necessary in order to assess effects of voltage-controllable equipment on the system ability to meet the desired objectives and to understand their tradeoffs. The performance objectives studied are efficiency measured in terms of generation cost, reliability measured in terms of voltage deviations from the pre-specified ranges, and power delivery to large load areas, New York City (NYC) in particular. Similarly, we report up-to-date improvements measured in terms of transmission loss minimization and management of critical voltages from using Automatic Voltage Control (AVC) in the PJM system. We conclude that the proposed CVC-Enabled HV Smart Grid could bring significant benefits to all three systems and that it would be a timely implementation. However, since the CVC architectures are non-unique it is important to fully understand the implications of their design on the achievable efficiency and reliability enhancements.

Index Terms—Voltage Controllable T&D Equipment, On-Load Tap Changing Transformers (OLTCs), Phase Angle Regulators (PARs), Controllable Shunt Capacitors (SCs), Flexible AC Transmission Systems (FACTS), Automatic Voltage Regulators (AVRs), AC Extended Optimal Power Flow (AC XOPF), Loss Minimization, Managing Extreme Voltages (MXV), Coordinated Voltage Control (CVC)-Enabled HV Smart Grid.

I. INTRODUCTION

This paper is motivated by our basic observation that the on-line coordination of all controllable resources within any given EHV and HV transmission system could enable more flexible utilization of equipment than is currently practiced without endangering reliable service. We suggest that this is both necessary and feasible [1]. There also exists a real window of opportunity to utilize the on-line remote sensing and command infrastructures being deployed by the utilities under the umbrella of smart grids. Progress is being made in integrating the synchrophasor measurements into the Supervisory Control and Data Acquisition (SCADA) systems and this will

ultimately enable accurate real power, phase angles, reactive power and voltage state estimation; much work is under way toward deploying sensing and synchronized communications. This progress is instrumental to managing controllable T&D equipment on-line correctively and adaptively. However, this will require a systematic framework for adjusting set points of the key T&D and generation controllable equipment. An example of an early implementation of Automatic Voltage Control (AVC) can be found in France [3].

Given both industry needs and technological potential, we describe in this paper one approach to moving forward. In Section II we first introduce our concept of CVC-Enabled HV Smart Grid as a single complex optimization problem. In Section III we describe several complex questions underlying an effective CVC design and implementation. Both challenges and potential benefits from deploying CVC for power systems operated by the three large ISOs are illustrated in Sections IV, V and VI, respectively. We close by highlighting preliminary findings and recommendations for moving forward, and the remaining open questions.

II. BASIC CVC PROBLEM FORMULATION

Consider first a traditional vertically integrated utility in which the system operator, the transmission owner and generator owner are one entity. Today's operating practice generally ensures reliable service first, and then schedules the least expensive generation, second. Reliable service is based on performing extensive power flow analyses and contingency screening to define operating limits within which real power is dispatched. These limits are determined by combining the equipment limits with the analysis findings and are considered to be hard constraints as the generation dispatch is performed. They are typically determined as part of planning studies for what may be considered to be the worst cases, and are enforced in operations. Generally, the present practice has not been to consider on-line adjustments of controllable equipment when computing system-constrained operating limits. For example, acceptable voltage limits are the result of a power flow-based analysis which does not optimize the settings of controllable equipment. Similarly, economic generation dispatch is done by the system operator without optimizing the settings of AVRs, OLTCs, SCs and FACTS. This practice is conservative as it does not utilize the full ability of the grid to deliver the most desirable real power dispatch. It is important to recognize that the need for on-line adjustments of voltage-controllable equipment is not for efficiency reasons only. This

is also necessary to avoid reliability risks when attempting to manage highly variable power and demand. Notably, the increased deployment of not-directly-controllable generation, such as wind and solar, will put a higher burden on T&D equipment for ensuring the controllability of the system as a whole within reliable limits. There are already instances of high-voltage problems during the low-load hours. Also, often the first limits to delivering real power into large load pockets, such as NY City (NYC) and Connecticut, are due to voltage problems [2]. All three systems described in this paper also experience voltage-related limits to delivering power from electrically distant generators. Perhaps most challenging is the fact that the system limits, unlike equipment limits, are highly dependent on the generation and load patterns in addition to the equipment status. This makes it practically impossible to pre-compute system limits at the planning stage. Instead, we propose that it is necessary to rely on an on-line CVC for adjusting key controllable equipment as conditions vary. This would enable efficient and clean utilization of power generation to best serve consumers without creating unreliable power delivery. The CVC computes and implements the best settings for voltage-controllable T&D and/or generation equipment as real power generation is dispatched to supply forecast demand. This is done by selecting the right performance objective on-line and by computing the best control dispatch within the given equipment limits. In what follows we illustrate how the choice of performance objective, and the number and type of decision variables fundamentally change what the power grid can deliver. Our main objective in this paper is to demonstrate the potential benefits from coordinated optimization of:

- controllable T&D equipment such as OLTCs, SCs, FACTS, and PARs (TO-owned controllable equipment);
- set points of automatic voltage regulators (AVRs) of generators (GO-owned voltage controllable equipment); and,
- real power generation (GO-owned).

III. THE CHOICE OF PERFORMANCE OBJECTIVES FOR A CVC-ENABLED HV SMART GRID

An important aspect of the proposed CVC concerns the choice of performance objectives when optimizing different controllable equipment. We consider in this paper that the key objective is reliable service, and efficient utilization is enabled to the extent possible. This is the least radical departure from today's operating practice. The major enhancement comes from not treating voltage-problem limits as hard constraints. These are, instead, the result of optimizing controllable equipment to ensure that hardware limits are observed. Potential benefits from relaxing these system limits are illustrated later in this paper for both the NE and NY power systems.

Any real power generation dispatch objective can be in principle enhanced by the CVC. In this paper we illustrate the effect of our proposed CVC on generation dispatch for minimizing O&M generation cost and for maximizing load-ability to NYC, using the NE and NY systems, respectively.

Dispatching real power objectives must be done within the specified hardware limits.

To ensure system operations within reliable voltage limits, the CVC-Enabled HV Smart Grid is implemented according to the following key performance objectives in the decreasing order of priority:

- Manage extreme voltages (MXV) to within the safe hardware voltage constraints. The industry practice is 0.98 pu-1.02 pu for normal conditions and 0.95 pu-1.05 pu during equipment failures. An exception is made for the voltage settings for nuclear plants, in particular, which are often outside these limits. This is done for the forecast generation and demand.
- Dispatch real power generation for efficiency within the limits obtained as a result of the first MXV step.

A. CVC-Protocols for Supporting Allocation of Responsibilities

The proposed CVC can be implemented either by a single centralized system operator, as implemented in PJM, or by a single system operator in cooperation with the owners of controllable equipment. In particular, the objective of managing extreme voltages to within safe hardware constraints can be done by a system operator simultaneously optimizing all controllable resources, or, by the TOs and GOs optimizing settings of their controllable equipment for a generation and demand profile given by the system operator. This could then be followed by the ISO scheduling real power generation given these (sub-optimal) T&D equipment settings, and then passing on the desired generation schedule to the TOs to optimize the settings of their own equipment. An interactive protocol by which all entities adjust their own controllable equipment, while exchanging the necessary information with the ISO may be a pragmatic approach since the TOs and GOs often have a much better understanding of the status of their equipment, particularly at lower sub-transmission and distribution levels, than does the HV grid system operator. According to this protocol, the TOs and GOs would retain the right to operate and control their own equipment. In principle several possible formal decompositions of a single optimization problem are possible. Depending on which decomposition is made, the information exchange required is different and the outcomes are different. In what follows we illustrate the outcomes of some possible CVC versions and the dependence of their performance on the choice of performance objectives and combination of controllable equipment. We illustrate results of both centralized and multi-layered CVC architecture for the case of NE system later in this paper.

B. From preventive to on-line corrective management of resources

In today's operations often the key limits to more efficient generation utilization during normal conditions come from imposing a preventive approach to real power dispatch. Real power generation is dispatched within the system limits which must be observed as though the worst contingency were to

take place. This is to be done without relying on dispatch of set points for voltage-controllable equipment. We propose that for most of the contingencies which are voltage-problem-limited it would be possible to dispatch real power generation more efficiently provided that voltage controllable equipment is adjusted when the contingency takes place.¹ The most measurable benefits from the proposed CVC would come from moving to a corrective management of resources during the contingencies, to the extent possible. There have been estimates of potential benefits reported up to 25% of total O&M generation cost [2]. The key idea of our proposed CVC-Enabled HV Smart Grid is to monitor conditions on-line, and to adjust all equipment that is available. The approach makes sense for both normal and contingency conditions as long as there is sufficient time to act before major problems occur. As a matter of fact, a continuous adjustment of resources as conditions vary is likely to lead to both more reliable and efficient operations when compared with today's preventive approach. We illustrate this claim later in this paper. Utilizing well-designed software for finding the best adjustments of controllable equipment to meet this objective is key to the successful implementation. In what follows we have used an AC XOPF to simulate and illustrate potential CVC benefits for NE and NY systems. A different AC OPF software was used for demonstrating potential benefits from adjusting set points of AVR, OLTCs and SCs for the PJM system.

IV. ILLUSTRATION OF THE PROPOSED CVC FOR THE CASE OF NEW ENGLAND POWER SYSTEM

We have simulated the NE system using a Summer Peak 2005 planning base case for normal conditions and for a severe (N-2) contingency. To begin with, the base case has many voltages which are outside 0.98 pu - 1.02 pu range. We refer to these as the "outliers". In what follows, we first simulate the proposed CVC as though all information is known to the ISO-NE. A centralized CVC is implemented to show: (1) the potential CVC efficiency effects seen in reduced generation cost, during both normal conditions and for the worst-case contingency; and (b) the potential CVC reliability enhancements seen in managing voltages close to the desired 0.98 pu - 1.02 pu range. We then assess the generation cost increase related to maintaining voltages within this "most reliable range". Finally, we propose a possible multi-layered implementation of the proposed CVC method and specify the basic binding information exchange protocol for allocation of responsibilities between the ISO-NE and equipment owners necessary to enable near-optimal performance when compared with the centralized CVC. We compare the performance of the centralized and this multi-layered CVC architectures.

¹The real exceptions to this statement are contingencies which would destabilize system dynamics; these limits must be observed unless there is sufficient fast control to stabilize the system. The stability-related limits are assumed to be included as hard limits in the proposed CVC.

A. The effects of CVC on generation cost in NE: Normal operation within voltage limits determined by base-case power flow

To illustrate the potential effects of implementing the proposed CVC on real power generation cost, we first allow voltages to vary within the ranges determined by the base case power flow. For example, if the base-case voltage at some bus was 0.95 pu, the permissible range of voltages at this bus would be between 0.95 pu and 1.02 pu. Similarly, if some other bus has a high voltage of 1.04 pu in the base case power flow, its permissible range would be 0.98 pu-1.04 pu. Shown as a black line of data points in Figure 1 are the base case voltages at all buses arranged from the lowest to the highest. The permissible low-voltage limit is then minimum of these voltages and 0.98 pu on bus-by-bus basis. Similarly, the permissible limit for high voltages would be maximum of these voltages and 1.02 pu, again on a bus-by-bus basis. Shown in Table I are the economic dispatch results when optimizing a subset of voltage-controllable equipment in addition to real power generation. The O&M generation costs range between \$90/MWh to \$0/MWh depending on fuel cost and heat rate. It can be seen in this table that the cost of generation given in the base case is \$688,092. When only the real power P_G is optimized and the generator voltages V_G and the taps of the OLTCs and SCs are as given in the base case, the optimal achievable generation cost is \$612,669. When both the real power P_G and the generator voltages V_G are optimized, the resulting generation cost is \$605,135. The least-cost generation is achieved when all controllable equipment is dispatched, namely real power P_G , generator voltages V_G and the taps of the OLTCs and shunt capacitors, and is \$604,391. The largest efficiency improvement possible is about 2% relative to the basic real power generation P_G dispatch with voltages as in the base case power flow. This is a significant annual saving.

B. The effects of CVC on generation cost in NE: The worst-case (N-2) contingency within voltage limits determined by base case power flow

Next, the worst-case (N-2) contingency is introduced into the base case, and economic dispatch is performed to assess the potential of managing resources efficiently and reliably during outages. Shown in Table I in the last row are the economic dispatch outcomes with different combinations of voltage controllable equipment. Notably, it is not possible to operate the system without adjusting AVR voltage settings of certain key generators. It can be seen that re-dispatch of generator voltages is key to the reliable operations during this critical contingency. The adjustments of taps are insufficient to ensure a reliable solution. Moreover, the taps do not contribute significantly to the overall cost reduction during this contingency. Finally, as expected, the generation cost of operating the system during the worst-case contingency is considerably higher than during normal conditions.

	No control	P_G	P_G, V_G	$P_G, Taps$	$P_G, V_G, Taps$
Base Case	\$ 688,092	—	—	—	—
Normal Operation	—	\$ 612,669	\$ 605,135	\$ 606,712	\$ 604,391
Worst Case ($N-2$) Contingency	—	not feasible	\$ 618,731	not feasible	\$ 614,233

TABLE I
ECONOMIC DISPATCH OUTCOMES AS A RESULT OF VOLTAGE OPTIMIZATION USED

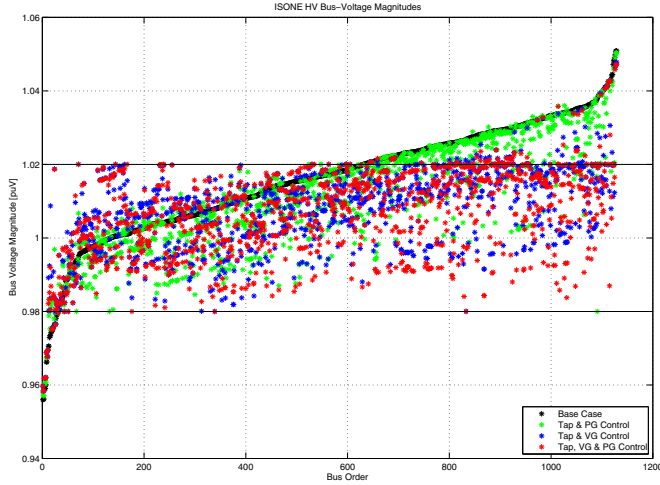


Fig. 1. Dependence of Voltage Profile on the Type of Controllable Equipment Following MXV Runs in HV NE Power System

C. The effects of CVC on enhanced reliability: Managing extreme voltages (MXV)

An AC XOPF is used to show the effects of controllable equipment on enhanced reliability. The reliability metric used by the AC XOPF as the performance objective to be minimized is the sum of extreme absolute-value voltage deviations over all HV buses in the NE system outside the desired 0.98 pu - 1.02 pu range. The results of simulating different combinations of controllable equipment on MXV are shown in Figure 1. It can be seen in this figure how the base case voltages (black line) can be brought closer to within the desired voltage range by adjusting: (1) generated power P_G and taps of OLTCs and SCs (green dots); (2) settings of AVR of generators and taps (blue dots); or (3) by adjusting all controllable equipment simultaneously (red dots). Observe that adjusting voltage set points of AVR on generators is most powerful since this action brings almost all voltage outliers to within 0.98 pu - 1.02 pu. As a general conclusion, these simulations show significant potential benefits from adjusting voltage controllable equipment on enhanced reliability.

D. The best possible CVC performance: Centralized architecture

Assuming first that the NE-ISO has a full information about the status of system equipment and full control of all voltage controllable equipment (both generation and T&D), we consider the “best” possible system performance when

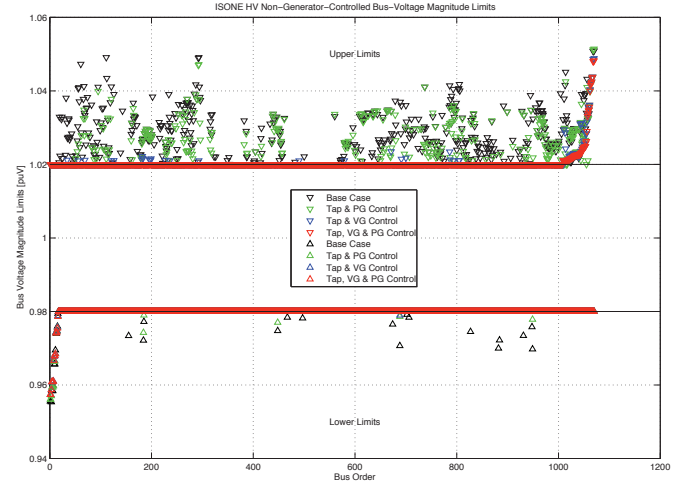


Fig. 2. Best possible ranges of permissible voltages obtained using MXV

all controllable equipment is adjusted as real power is being dispatched. As discussed above, we propose a CVC which gives the highest priority to reliability. This means that the “best” solution would be the one which has minimized the deviations of voltage outliers from their desired ranges. The result of this MXV optimization with respect to all controllers ($P_G, V_G, Taps$) is the best possible range of permissible voltages, controlled and uncontrolled. This range is defined by assigning the range of permissible voltage at each bus as: (a) 0.98 pu - 1.02 pu if the solution of MXV is within this range; (b) between the MXV voltage and 0.98 pu and 1.02 pu; (c) between the MXV voltage and 0.98 pu for buses whose MXV voltages are above 1.02 pu. Shown in Figure 2 are the ranges of reliable voltages obtained this way for different combinations of controllable equipment.

Next, an economic dispatch optimization is carried out with respect to all controls ($P_G, V_G, Taps$) with the objective of minimizing total generation cost within the best possible voltage limits determined in the first MXV step. For the case of NE system the total generation cost obtained after the second optimization is \$643,848. We refer to this result as the “most reliable” economic dispatch. Notably, when comparing this generation cost to the economic dispatch result \$604,391 shown in Table I it is concluded that the “cost of maintaining the most reliable voltages” is about 6% of total economic dispatch cost. The latter optimization uses the wide voltage limits defined by the base case voltage, namely the black dots in Figure 1. This result points out to the need for defining the

hard equipment reliability constraints as carefully as possible as they have major impact on economic efficiency.

E. A possible multi-layered CVC implementation

We finally consider the question of protocols for the allocation of responsibilities and the corresponding information exchange. We propose an interactive three-layer protocol for implementing a near-best-possible CVC while allowing the owners of voltage controllable equipment to make decisions regarding voltage ranges within which the ISO-NE performs its economic dispatch. The three steps are as follows.

- ISO-NE sends to the owners of voltage controllable equipment anticipated power demand and generation.
- The equipment owners perform MXV with respect to the controls (V_G , $Taps$) only. This optimization results in the acceptable voltage ranges bounded by upper and lower triangle symbols in Figure 2. The controlled equipment is frozen at the optimal values. The range of acceptable voltages and the optimized (V_G , $Taps$) are passed on to the ISO.
- The ISO-NE optimizes its real power generation P_G to minimize the total generation cost within the ranges of voltages and for the optimized (V_G , $Taps$) given by the equipment owners.

The economic dispatch obtained using this three-layer information exchange after one iteration is \$645,482 and is very close to the cost of the most reliable economic dispatch described above, \$643,848. This simulation represents proof-of-concept that a near-optimal multi-layer implementation would be possible by which the owners of controllable equipment decide most reliable voltage ranges and the NE-ISO optimizes the total generation cost within those ranges.

V. POTENTIAL CVC BENEFITS IN THE NEW YORK POWER SYSTEM

In this section we summarize our findings from the recently performed study using AC XOPF for the NY power system. In this study we have assessed existing voltage-related limits and explored potential benefits from coordinating management of controllable equipment [2]. Of particular interest is the potential increase of power delivered to NYC from Canada.

Today's procedures by the NYISO do not utilize on-line adjustments of voltage-controllable equipment. In our study one of the main objectives was to understand how would the results of economic dispatch and ability to deliver power to large loads change if the controllable equipment were to adjust during both normal and contingency conditions. Most generally, we have learned that the three objectives of (1) long-range power transfer across NYCA, (2) maximizing flows across specific interfaces inside NYCA, and (3) fair economic dispatch throughout NYCA often conflict, and are best served with control dispatches specific to each objective. For example, the greatest possible large power transfer from Canada to NYC does not result in the maximum possible flow across many of the interfaces along the way. Similarly, economic dispatch does not result in the stress of any single interface over another.

Contingency?	Voltage Range	Thermal Limit	Generation Cost [\$/Hr.]	Generation Cost Increase
No	0.98-1.02	Rate A (Normal)	1,110,290	Benchmark
Yes	0.98-1.02	Rate A	1,145,554	3.2%
Yes	0.95-1.05	Rate A	1,120,197	0.9%
Yes	0.98-1.02	Rate B (LTE)	1,114,792	0.4%
Yes	0.95-1.05	Rate B	1,080,022	-2.7%

Fig. 3. Effects of Voltage Limits on Generation Cost During A Severe Contingency

A. Critical constraints

We have found that there exists a set of constraints, including voltage limits and thermal line-flow limits, which are binding in many of the optimizations considered in this study. These constraints likely identify locations at which the system should be physically improved, locations at which system operation should be closely monitored, or locations at which corrective actions during a contingency will be important. Interestingly, the critical locations are not always at the highest voltage buses. Important binding constraints are observed in equipment operating down to 115 kV. Shown in Figure 3 are the effects of varying voltage limits on generation cost during a severe (N-1) contingency in this system. It can be seen that these limits affect the generation cost in a sensible but not necessarily predictable way. Nevertheless, it is important to have robust software that can optimize the cost as system status and limits vary in order to achieve the greatest possible savings.

Optimal voltage in support of economic dispatch: When performing economic dispatch, voltages are maintained between 0.98 pu and 1.02 pu in runs without contingencies, and between 0.95 pu and 1.05 pu when simulating critical contingencies, unless the original voltage was outside these limits. It can be seen in Figure 4 that economic dispatch cost minimization during normal conditions (no contingency) requires a 3% increase in voltage in Area E, 1% increase voltage in Area F, 1% voltage increase in Area C, and 2% voltage decrease in Area J. In this specific run this is achieved by a combination of taps of OLTCs and AVR set points of generators.

B. Comparison of NYC Area Imports

Figure 5 summarizes some of the loadability results, and compares the load served in the NYC Area to that achieved with other optimizations. For three sets of optimizations, the table compares the resulting total load served in the NYC Area (NYISO Areas 9-11), referred to here as "NYC" and the interface flow associated with the optimization. One set focuses on the Dysinger-East (DE) and West-Central (WC) interfaces, one set focuses on the Central-East (CE) interface, and one set focuses on the Total-East (TE) interface. These sets

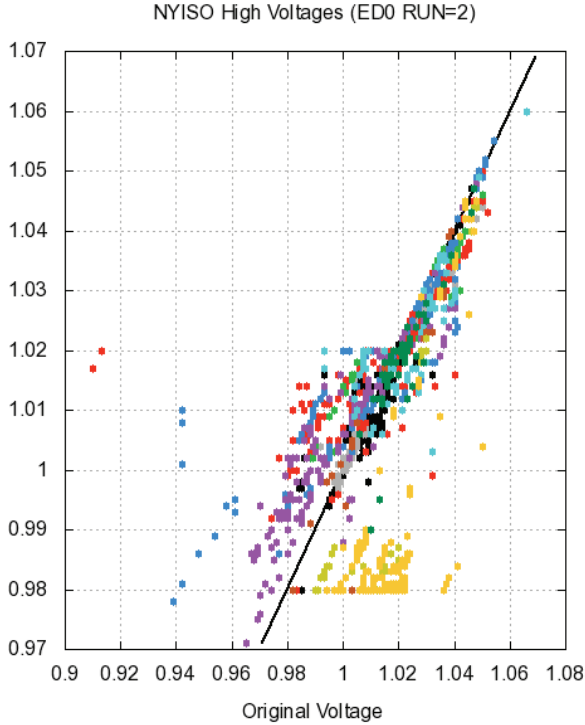


Fig. 4. Comparison of the Base Case and Optimized Voltage by Adjusting AVRs for economic Dispatch

Run Set		Run 0		Run 2		Run 7	
		NYC Load [MW]	Int. Flow [MW]	NYC Load [MW]	Int. Flow [MW]	NYC Load [MW]	Int. Flow [MW]
DE/WC	L2	21156	DE: 2285 WC: 966	21542	DE: 2798 WC: 1438	21594	DE: 2800 WC: 1441
	PP[DEWC]3	20122	DE: 2154 WC: 846	20175	DE: 2203 WC: 900	20242	DE: 2203 WC: 902
	IF[DE]1	20546	2388	20867	2812	20553	2812
	IF[WC]1	20581	1059	21016	1453	20630	1457
CE	L5	21381	3109	21509	3207	21538	3209
	PP[CE]2	20762	3027	21114	3248	21332	3292
	IF[CE]1	21116	3398	21235	3482	21193	3515

Fig. 5. A comparison of NYC Load and Interface Flows Across Various Optimizations

of optimizations are chosen for comparison because, within each set, the sending areas, the receiving areas, the study areas, and the external constraints are approximately the same. Additionally, the applied contingencies are identical within each set. Each optimization set includes a loadability optimization (L) that seeks to maximize the load served in New York City, a points-to-points optimization (PP) that seeks to maximize the source and sink injections in the sending and receiving areas, and an interface optimization (IF) that seeks to maximize the flow through the associated interface. Note that there are actually two independent IF optimizations in the case of the Dysinger-East and West-Central set. As should be expected, within each set of optimizations, the loadability optimization always yields the greatest load served in NYC, while the interface flow optimization always yields the greatest interface

flow. Interpreted differently, the maximum load served in NYC is not achieved by maximizing any given interface flow, and maximizing a given interface flow using sinks in NYC does not result in the maximum load served.

Figure 5 shows that neither maximizing points-to-points transfers nor maximizing interface flows should be used to maximize the load served in NYISO Areas 9-11 even during the same critical contingency. Rather, a loadability optimization should be used; the main difference between the three is the optimization objective. For example, consider the three optimizations involving the Total-East interface. In this case the loadability optimization consistently results in 200 MW or more load served in NYC, yet it does so with the Total-East interface flow that is about 100 MW less than that achieved with the interface flow optimization. The interface flow optimization achieves the highest interface flows. Comparable load and interface flow differences are observed in the case of Central-East, while the differences are even greater in the case of Dysinger-East/West-Central. Comparing the results across contingencies it is apparent that the Total-East contingency presents the greatest limit to delivering power to NYC over long distances across NYCA. Finally, comparison of the optimization shows that the best optimization is one that matches the objectives of any given task whether the task is serving loads or maximizing interface flows.

C. Optimal voltage in support of NYC loadability

When attempting to maximize possible loadability in the areas of interest, optimized voltages in Areas E, F, A, and B are higher by 4%, 1%, 0.5% and 0.1% relative to the base case. Optimized voltages in Area C are scattered and high at some points for the case without contingencies; see Figure 6.

VI. POTENTIAL BENEFITS AND CHALLENGES OF AUTOMATIC VOLTAGE CONTROL (AVC) IN PJM

PJM transmission system has the generation capacity of 174,000 MW, and the peak load of 158,000 MW. With the increase of PJM power grids scale, voltage control and Var management become more and more challenging. So far each TO in the PJM system determines their own area voltage schedule based on their individual assessments. If a TO has no voltage schedule for its area, PJM defines a default one. This suboptimal approach is no longer suitable for the higher security and economy requirements of the PJM system operations. Because of this, recently a hierarchical voltage control architecture [4]-[5] was developed for PJM Optimal Voltage Control (AVC) system, which is based on the concepts of optimization with zone division and pilot buses. The AVC system consists of Data I/O Management, Control Strategy Calculation, Operation Log Management and Control Process Dispatch. It is interfaced with PJM EMS system, automatically receiving real-time snapshots from PJM EMS on a periodical basis.

A. Overview of PJM AVC System

The overview architecture of implemented AVC system is shown in Figure 7. The AVC system is interfaced with PJM

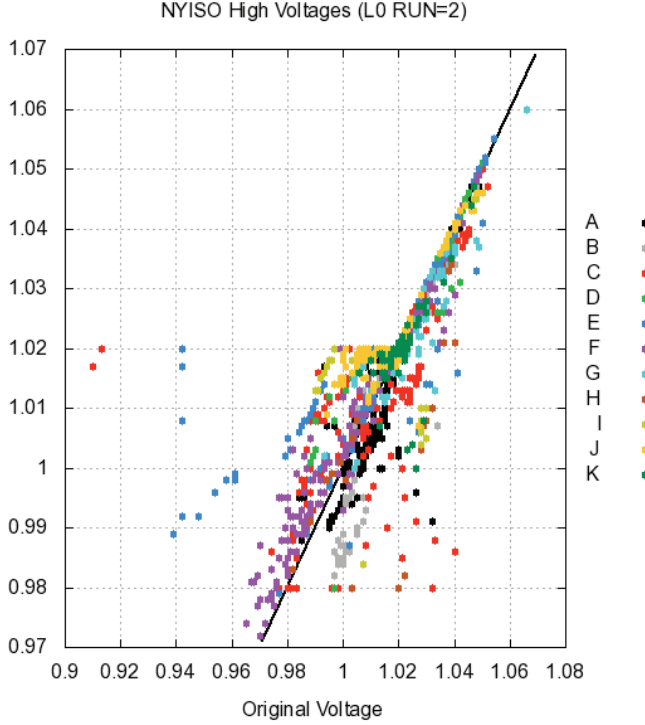


Fig. 6. Comparison of base case and optimized voltages for NYC loadability

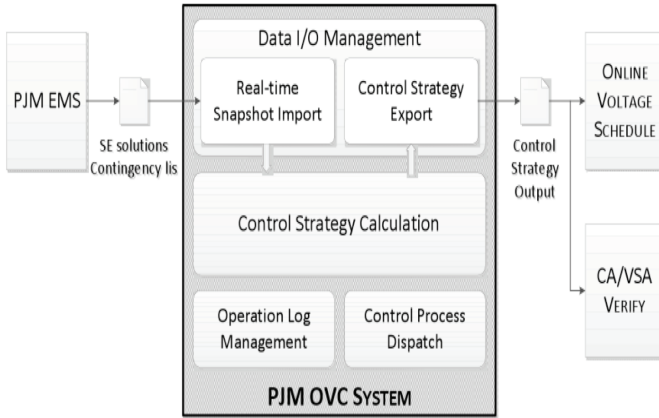


Fig. 7. Architecture of AVC demonstration system

EMS to receive real-time snapshots including State Estimation (SE) solutions with real time network models and a contingency list from Security Analysis (SA) periodically. With those two inputs, AVC system carries out the optimization of control strategy calculation and outputs optimal voltage schedule for each generator, OLTC and switchable SCs within the PJM system.

AVC system is a centralized control system, which can coordinate the actions of all voltage/VAR control devices across the whole power grid. The control strategy calculation module in it is responsible for the optimal control decision-making for the entire power grid.

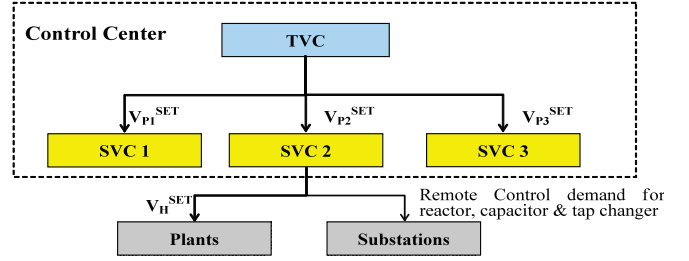


Fig. 8. Hierarchical structure of control strategy calculation module

Logically the control strategy calculation module consists of two levels, i.e. the secondary voltage control (SVC) and Tertiary Voltage Control (TVC). The logical hierarchical structure was shown in Figure 8. TVC offers the optimal set-points of the pilot bus voltages to SVC, and then the latter coordinates all generators to keep the voltages at the pilot buses on their optimal reference values. Control strategy calculation adopts the reactive optimal power flow (ROPF) with the goal of system loss minimization to obtain the optimal set-points of the pilot bus voltages. In ROPF calculation model, reactive outputs of all generators inside of PJM system are considered as control variables. Security constraints are considered in high dimensions as well, which include all voltage constraints and VAR constraints inside the PJM system. Since control strategy presented from AVC should meet requirements of Contingency Analysis (CA), CA results were used in the voltage constraints on critical buses. If the two objectives conflict each other, control strategy calculation will make a trade-off between them by readjusting the voltage constraints on critical buses.

B. Results and conclusions

PJM AVC system has been run on-line parallel with the existing voltage control system at PJM control center since January 2010. PJM operation engineers performed the study and analysis by comparing the optimal solutions and existing solutions for each interval using software with load flow, contingency assessment and voltage stability assessment. The results in this paper are from June to September in 2010.

The average voltage before and after AVC calculations are compared with voltage increases about 0.004 p.u. by AVC. The comparisons of average system losses for all the four months with loss reduction of about 1.19% for each snapshot by AVC. The average VAR reserve under AVC increased in every month, with an average increase in reserve of 1.08% over four months.

The comparisons of average base voltage limit violation numbers for all the four months with an average reduction of 9.3 violations, about 99.3% snapshots were better. For evaluating AVC effects on voltage stability margin, the

Time	EAST	CENTRAL	WEST	BED-BLA	APSOUTH
Jun	5.9%	9.0%	12.2 %	6.3%	16.0%
Jul	5.2%	8.2%	12.5%	7.1%	14.4%
Aug	8.5%	9.4%	14.8%	8.4%	14.8%
Sep	10.7%	13.4%	15.8%	8.9%	19.1%

Fig. 9. Average Voltage Stability Margins Increase (%)

voltage stability margin of five typical interfaces, “EAST”, “CENTRAL”, “WEST”, “BED-BLA” and “APSOUTH”, are calculated before and after AVC. The voltage stability margins after AVC are greatly improved. The percentage increases for the average voltage stability margins are shown in Figure 9.

APSOUTH interface is used as an example for explaining why the AVC optimal voltage solutions give larger voltage stability margins are greatly improved for the following reasons: (1) The more reasonable distribution of reactive power; (2) The Var reserve in Area13/Zone16 becomes less after AVC, and the voltage becomes higher, that means this area can meet the reactive power needs by itself, and the transmission of reactive power between different areas is reduced; (3) The more flat voltage distribution the mean square deviation of voltage is much smaller after AVC, that means the bus voltages distribution is more flat; (4) The higher voltage level; (5) The bus voltages related to APSOUTH interface become much higher after AVC, especially for receiving side buses. The experience of the AVC system running on-line for PJM real time operations has shown the significant potential benefits in PJM system in both aspects of security and economy: (1) Average reduction of transmission system loss in PJM system is about 1.19%, which means energy savings of more than 220 million kWh/year and monetary savings of more than \$ 17 million/year; (2) Average increase of system VAR reserve is about 1.08% in PJM; and, (3) Improvement of PJM system security for pre-contingency and post-contingency is significant, especially for the voltage stability. That means reducing the number of constraints (congestion) in PJM transmission system, which will be reflected in the PJM energy market through the reduction of congestion cost.

VII. PRELIMINARY CONCLUSIONS AND OPEN QUESTIONS

In this paper we introduce the concept of a Coordinated Voltage Control (CVC)-enabled HV Smart Grid. The CVC architectures can vary depending on the control equipment used, protocols for information exchange and on the allocation of responsibilities between the ISOs and the owners of voltage controllable equipment (TOs, GOs and, ultimately Distribution Owners (DOs)). We simulate NE, NY and PJM systems with and without CVC to illustrate possible enhanced system operation. Potential benefits measured in terms of more

efficient real power utilization within the reliable equipment limits are described. We first simulate and analyze the role of CVC in support of reliable least-cost generation dispatch for the NE and NY systems. We then illustrate how the same CVC architecture could be used to support power delivery reliably and efficiently to NYC. Notably, we show that CVC would make it possible to operate the system reliably during many contingencies during which this is not possible using today’s operating practice. Last but not least, we describe the centralized Automatic Voltage Control (AVC) architecture currently implemented in PJM and report the estimated benefits up to date for this system. The objective of AVC is on-line optimization of voltage controllable equipment for minimizing transmission loss during normal conditions. The same AVC architecture is used to adjust voltage controlled equipment for implementing corrective actions during contingencies and for dynamically relaxing key PJM transfer limits within the reliable voltage limits. While the AVC architecture may appear to be a different concept from the CVC concept illustrated for NE and NY, it can be considered broadly as yet another design of a CVC-Enabled Smart Grid. PJM is using AVC as an advisory tool and the next step would be its integration in on-line operations. Based on the simulations and findings for these three systems we propose that a CVC-enabled HV Smart Grid could bring tremendous benefits to both reliability and enhanced generation utilization and better service to the end users. We recommend that some version of a CVC-enabled HV Smart Grid should be considered as part of a HV Smart Grid design. The proposed CVC gives priority to reliability over efficiency. Questions concerning incentives to the participants in the proposed CVC-enabled HV Smart Grid should be subject of future work. The solutions should be based on formal problem posing for designing incentives for ISOs, TOs and GOs to become an active participant in the CVC-Enabled Smart Grid.

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